



## 1. Project Overview

A simple Kotlin Retail application designed as a digital storefront for the Shopistry Bakery Business. It offers a role-based platform where customers can login into their registered accounts, browse the bakery's catalogue & order deliveries of their chosen pastries & desserts, while admins can manage the shop's inventory & the information of each product.

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## 2. Objectives & Scope

### 2.1. Objectives

- Provide a digital storefront for Shopistry customers to browse & transact with the business online.
- Provide a role based platform where admins can manage the store's inventory & product information remotely, and customers can create accounts to keep track of info, orders and interact with the Bakery.

### 2.2. Scope

**Shopistry** will cover:

- **Digital Storefront:** Browse items, view details & stock, add items to cart, place order & delivery. Filters: Sort by category, Sort by Price, Search by Name.
- **User Accounts:** Register, login, logout, customer & admin roles, set profile picture, edit profile info (name, email, password).
- **Admin Dashboard:** Add/Remove items, update stock count, modify item info (name, price, description, category, picture)

- **Cart System:** Add & remove items from cart, view items in cart, checkout & place order.
- **Modern GUI:** Minimalist Theme, Intuitive layout, responsive buttons and fields.

### 2.3. Out of Scope

- Digital Payment System (only Cash On Delivery available)
- Working order system (Current version features a mock Order System)

## 3. Release Plan

| Release Version                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Duration | Objectives                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------|
| Release 1.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Ongoing  | To release a prototype for basic feature testing of Shopistry |
| Key Deliverables                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |                                                               |
| <ul style="list-style-type: none"> <li>• <b>Admin Dashboard</b><br/>Manage stock &amp; item info. Add/remove items.</li> <li>• <b>Customer Dashboard</b><br/>Browse bakery catalogue via scrollable grid. Details popup for each item when clicked. Add to cart/view cart buttons. Product search bar. Filter buttons: sort by price, sort by category.</li> <li>• <b>Navigation Bar</b><br/>Navigate to Dashboard, navigate to Cart, navigate to Profile</li> <li>• <b>Cart Page</b><br/>Display items in cart, remove items from cart, checkout &amp; place order button. Payment method display (currently limited to Cash On Delivery).</li> <li>• <b>Account Registration</b><br/>Input name, email &amp; password. Checkbox to register account as Administrator. Input validation for email &amp; password.</li> </ul> |          |                                                               |

- **Profile Page**

Displays account information, edit account information, set account Profile Picture.

- **Theme / UI**

Modern minimalist interface.

### 3.1. Deployment Environment

#### Staging:

- Run locally on developer devices or emulators on the IDE [Android Studio] for testing.
- GitHub repository for version control and code review.

#### Production:

- Full packaged .apk installed directly from developer machine to user phone.
- Runs on Android Devices with SDK Version 36+, minimum SDK 33.
- Dependencies: Kotlin 2.3.20, Firebase

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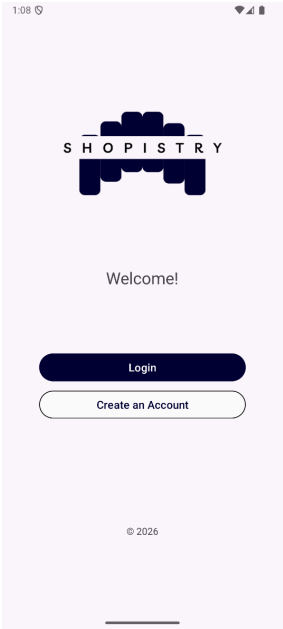
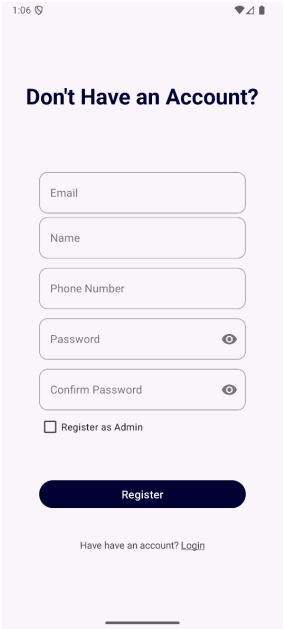
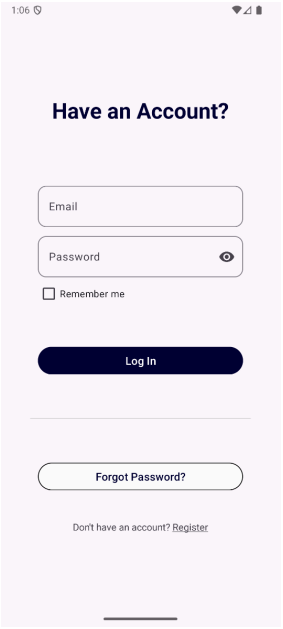
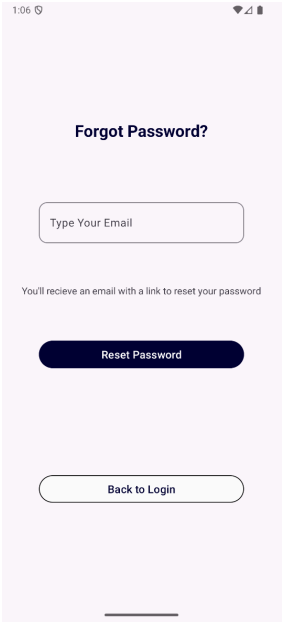
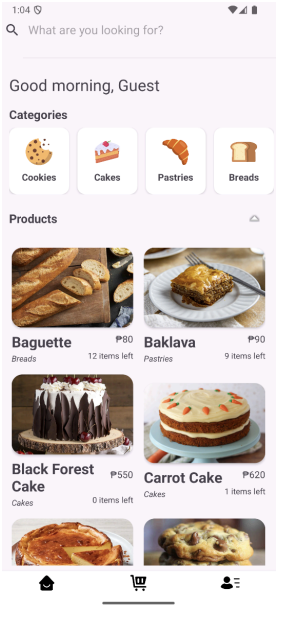
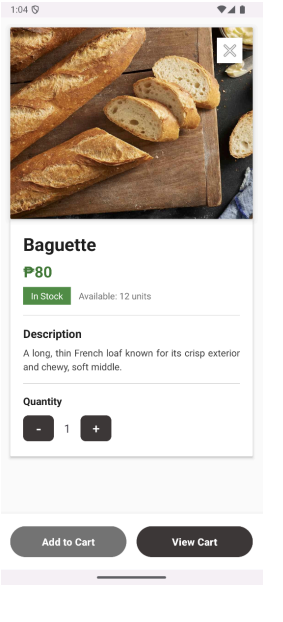
## 4. Environment & Dependencies

| Category    | Technology/Tool            | Purpose                                         |
|-------------|----------------------------|-------------------------------------------------|
| Environment | Kotlin version 2.3.20      | Language used to develop the program            |
| Frontend    | XML View System            | To design & generate the UI for the application |
| Backend     | Firebase Realtime Database | To store & manage the application's database    |

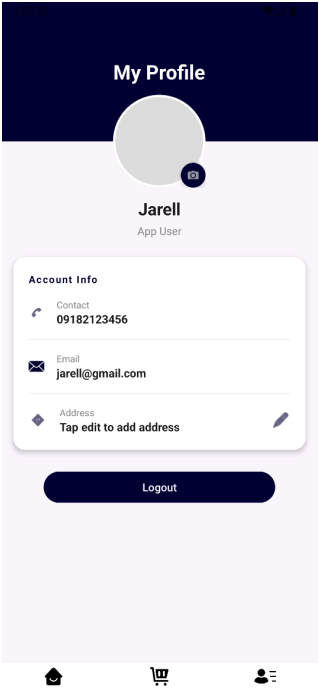
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|----------------------------|--------------------|--------------------------------------------------------------|
| <b>Image Adapter</b>       | Glide              | To process & display images in the application through links |
| <b>User Authentication</b> | FirebaseAuth       | To manage user account creation and authentication           |
| <b>Project Management</b>  | Facebook Messenger | Task tracking & planning                                     |
| <b>Version Control</b>     | Github             | For version control & organized collaborative development    |

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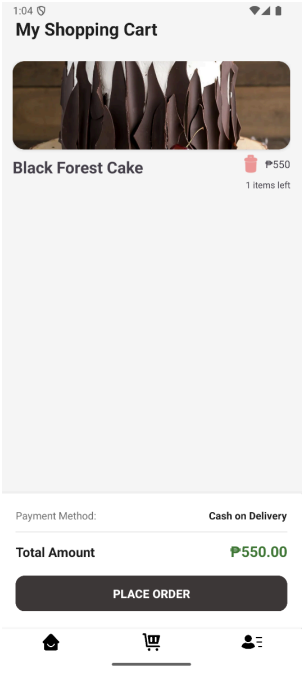
## 5. OEE Calculator Pro Snapshots

| Welcome Page                                                                                                                                                                                                                                                | Register Page                                                                                                                                                                                                                                                                                                                                                                        | Login Page                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>1:08</p> <p>SHOPISTRY</p> <p>Welcome!</p> <p>Login</p> <p>Create an Account</p> <p>© 2026</p>                                                                         |  <p>1:06</p> <p>Don't Have an Account?</p> <p>Email</p> <p>Name</p> <p>Phone Number</p> <p>Password</p> <p>Confirm Password</p> <p><input type="checkbox"/> Register as Admin</p> <p>Register</p> <p>Have have an account? Login</p>                                                               |  <p>1:06</p> <p>Have an Account?</p> <p>Email</p> <p>Password</p> <p><input type="checkbox"/> Remember me</p> <p>Log In</p> <p>Forgot Password?</p> <p>Don't have an account? Register</p>                                                     |
| Forgot Password Page                                                                                                                                                                                                                                        | Customer Dashboard                                                                                                                                                                                                                                                                                                                                                                   | Detailed Product View                                                                                                                                                                                                                                                                                                              |
|  <p>1:06</p> <p>Forgot Password?</p> <p>Type Your Email</p> <p>You'll recieve an email with a link to reset your password</p> <p>Reset Password</p> <p>Back to Login</p> |  <p>1:04</p> <p>What are you looking for?</p> <p>Good morning, Guest</p> <p>Categories</p> <p>Cookies Cakes Pastries Breads</p> <p>Products</p> <p>Baguette P80 12 items left</p> <p>Baklava P90 9 items left</p> <p>Black Forest Cake P550 0 items left</p> <p>Carrot Cake P620 1 items left</p> |  <p>1:04</p> <p>Baguette</p> <p>P80</p> <p>In Stock Available: 12 units</p> <p>Description</p> <p>A long, thin French loaf known for its crisp exterior and chewy, soft middle.</p> <p>Quantity</p> <p>- 1 +</p> <p>Add to Cart View Cart</p> |

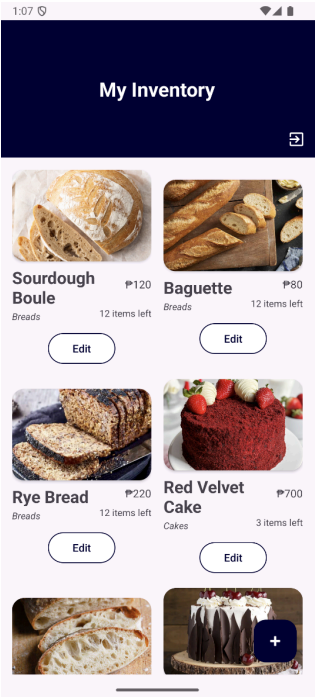
Profile Page



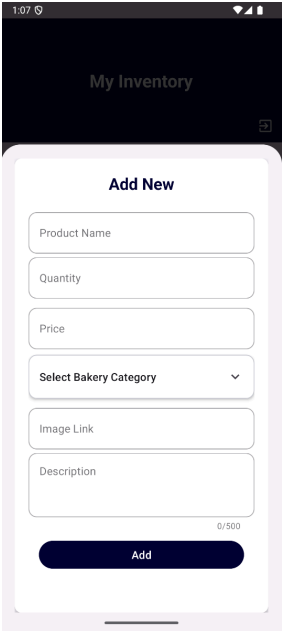
Cart



Admin Dashboard



Admin Add/Edit Product  
Popup



## 6. Summary / Conclusion

**Shopistry App** is a dedicated digital storefront and inventory management solution designed to streamline the connection between the Shopistry Bakery and its customers. Its minimalist interface, role-based access control, and real time synchronization through Firebase provide a seamless experience that bridges the gap between manual ordering and modern e-commerce convenience.

Through a robust XML-based View System and Kotlin integration, **Shopistry** ensures a responsive and intuitive environment for both administrators managing stock & customers browsing its catalogue. By leveraging Firebase for authentication and database management, the app provides a reliable, scalable foundation for future expansions, such as integrated payment systems and live order tracking, ultimately empowering the business to grow its digital presence with minimal technical overhead.